# S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

## Practical No. 10

**Aim: Apply prompt engineering to construct a prompt for learning the concept of Adversarial Search.**

**Name of Student:**

**Roll No.:**

**Semester/Year: V/III**

**Academic Session:**

**Date of Performance:**

**Date of Submission:**

**AIM:** To design and refine an effective prompt using prompt engineering techniques for learning the concept of Adversarial Search from Artificial Intelligence.

**OBJECTIVE/EXPECTED LEARNING OUTCOME:**

The objectives and expected learning outcome of this practical are:

- To be able to understand the concept of a *prompt* (input given to an AI model).
- To be able to understand the concept of *prompt engineering* (designing inputs to get desired outputs)
- To be able to work with basic idea and models like ChatGPT.
- To be able to distinguish between a vague prompt and a well-structured one.

**THEORY:**

## i. Fundamentals of Prompt Engineering

Prompt engineering is the process of designing and structuring inputs (prompts) to get accurate, relevant, and well-formatted outputs from AI systems such as ChatGPT. Prompt engineering is how you ask a question to AI so that it gives the best possible answer.

## ii. Characteristics of a Good Prompt

A good prompt is based on the following core principles:

- **Clarity** → Avoid ambiguity
- **Specificity** → Define exact task
- **Context** → Provide background
- **Constraints** → Limit output (length, format, style)
- **Intent** → Clearly state what is required

## Example:

- Weak: *"Explain AI"*
- Strong: *"Explain AI in 5 bullet points with real-world examples for beginners"*

## iii. Prompt Patterns / Techniques

The common patterns of the prompt along with their usage are as follows:

### a) Role-Based Prompting

"You are an AI professor…"

### b) Step-by-Step Prompting

"Explain step-by-step…"

### c) Few-shot Prompting

Provide examples in the prompt

### d) Instruction + Constraints

"Explain in 100 words using simple language"

### e) Chain-of-Thought (Guided Reasoning)

"Show reasoning before final answer"

## iv. Task-Oriented Prompting

Based on the tasks we can have different ways to provide the prompt. Following are the examples to design prompts for different tasks:

- **Learning/Explanation**
- **Coding**
- **Summarization**
- **Question generation**
- **Problem solving (AI, Cloud, etc.)**

## v. Iterative Refinement (Very Important)

The first prompt may not be the best one. So, we must learn how the prompt can be refined to get most appropriate outcome:

- First prompt ≠ best prompt
- Improve by:
  - Adding constraints
  - Asking follow-up questions
  - Correcting output

## Concept:

Prompt → Output → Refine → Better Output

## vi. Evaluation of AI Output

It is very important to evaluate the output generated by AI. Critical thinking should be done to find the answer to the following:

- Is the answer **correct**?
- Is it **complete**?
- Is it **biased or vague**?
- Does it follow instructions?

## vii. Ethical and Responsible Use

- Avoid Misuse and Plagiarism
  - Do not copy AI-generated content blindly for assignments or research
  - Always ensure originality and proper attribution when required
- Verify Information
  - AI tools like ChatGPT may generate incorrect or outdated information
  - Cross-check important facts before using them in academic or professional work
- Use AI as a Support Tool
  - AI should assist learning, not replace critical thinking
  - Students should understand concepts rather than depend entirely on AI-generated answers

## viii. Basic Applications in Academics

- Generate notes
- Create MCQs and scenario-based questions
- Debug code
- Explain complex topics simply

## ix. Perform the following tasks:

- Convert a bad prompt → good prompt
- Generate:
  - Notes
  - Quiz questions
  - Code snippets
- Design a prompt for:
  - Adversarial Search explanation

**EXPECTED OUTPUT:**

- Definition
- Game tree explanation
- Minimax working
- Alpha-Beta optimization
- Example
- Complexity analysis

**INPUT & OUTPUT:**

| Attempt | Prompt Quality | Output Quality | Improvements |
|---|---|---|---|
| 1(Basic) | "Explain Poker in AI." | Too vague, lacks structure, missing algorithms | Add specific topics (minimax, example, complexity) |
| 2(Improved) | "Explain Poker as an adversarial search problem. Include game tree, minimax, and example." | Better, but lacks clarity on uncertainty | Add constraints: diagrams, table, structured headings |
| 3(Refined) | "Explain Poker as an adversarial search problem in AI. Include definition, game tree, minimax, alpha-beta limitations, example with actions, and complexity analysis in structured format with tables." | High-quality, detailed, well-structured answer | Final optimized prompt |

**CONCLUSION:**

**DISCUSSION QUESTIONS:**

1. What is prompt engineering?

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2. Why is prompt refinement important?

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3. What is Alpha-Beta pruning?

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4. Difference between weak and strong prompts?

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**REFERENCES:**


- https://learnpromptengineering.dev
- https://www.linkedin.com/pulse/40-prompt-engineering-resources-upskilling-spencer-taylor-xje0e/
- https://justprompting.in/